lab14.py - C:/Users/Deekshith/AppData/Local/Programs/Python/Python314/lab14.py (3.14.2)

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```python
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
# Dataset  (you load csv data )
data = {
'Salary': [50000, 60000, 75000, 80000, 120000, 95000, 40000, 70000],
'Years_Experience': [1, 2, 3, 4, 5, 6, 1.5, 3.5],
'Experience_Level': ['Junior', 'Junior', 'Mid', 'Mid', 'Senior', 'Senior', 'Junior', 'Mid']   }
df = pd.DataFrame(data)

# 1) Salary Distribution
sns.histplot(df['Salary'], kde=True)
plt.title("Salary Distribution")
plt.show()
# 2) Pair Plot (Salary vs Years_Experience)
sns.pairplot(df[['Salary', 'Years_Experience']])
plt.show()
# 3) Correlation Heatmap
sns.heatmap(df[['Salary', 'Years_Experience']].corr(), annot=True, cmap='coolwarm')
plt.show()
# 4) Line Plot: Salary Trend by Years of Experience
sns.lineplot(x='Years_Experience', y='Salary', data=df, marker='o')      # 'o' - circle  's' (square), '^' (triangle), '*' (star) (don't type)
plt.title("Salary Trend by Experience")
plt.show()
# 5) Scatter Plot: Salary vs Years of Experience
sns.scatterplot(x='Years_Experience', y='Salary', data=df, s=100)
                                        # s = size of each scatter point.
plt.title("Salary vs Experience")
plt.show()
# 6) Bar Plot: Experience Level vs Salary
sns.barplot(x='Experience_Level', y='Salary', data=df)
plt.title("Salary by Experience Level")
plt.show()
```

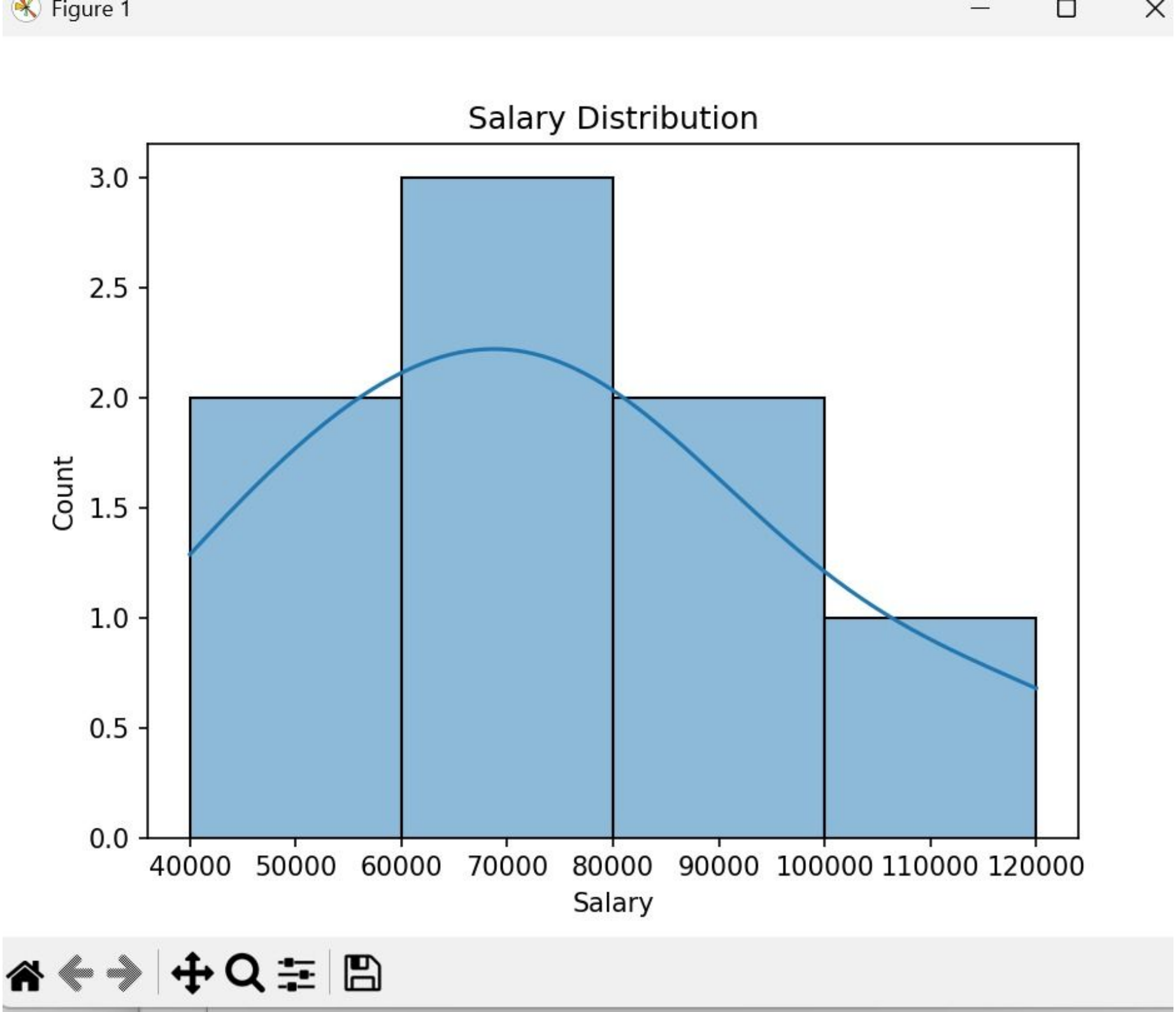
Figure 1
Salary Distribution
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Count
40000
50000
60000
70000
80000
90000
100000
110000
120000
Salary

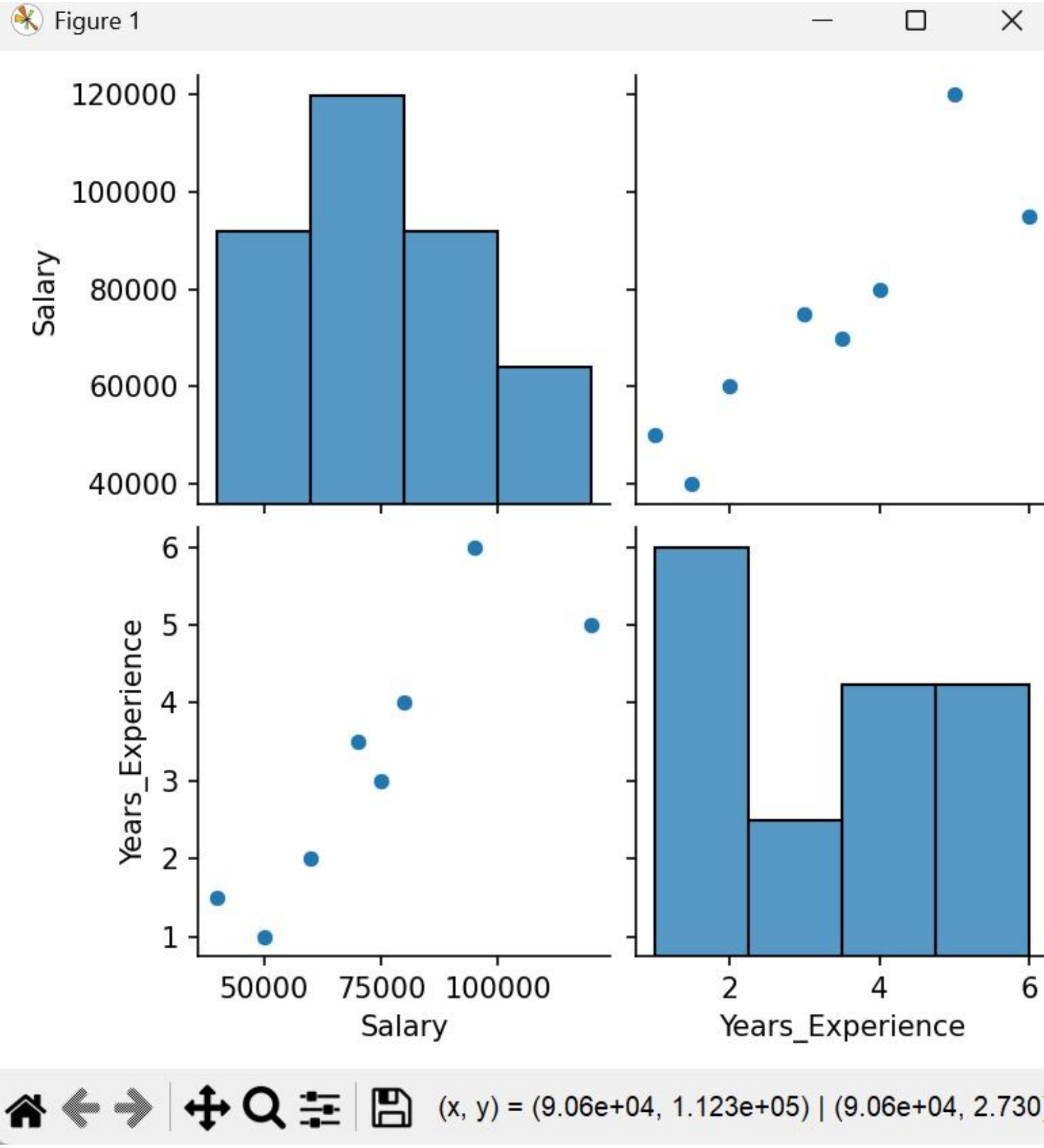
Figure 1
120000
100000
80000
60000
40000
Salary
6
5
4
3
2
1
Years_Experience
50000
75000
100000
Salary
2
4
6
Years_Experience
(x, y) = (9.06e+04, 1.123e+05) | (9.06e+04, 2.730

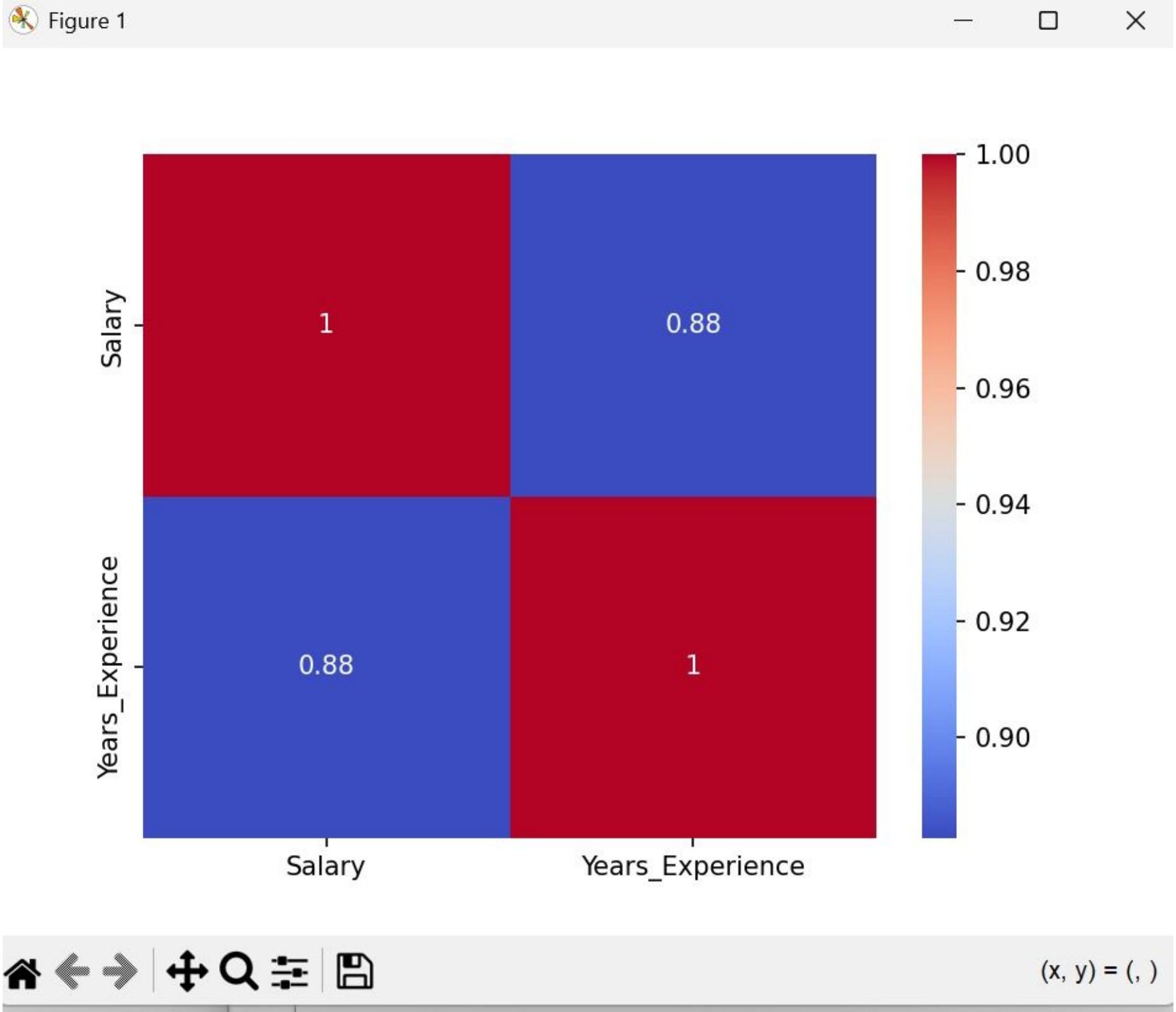
Figure 1
1
0.88
0.88
1
Salary
Years_Experience
Salary
Years_Experience
1.00
0.98
0.96
0.94
0.92
0.90
(x, y) = (,)

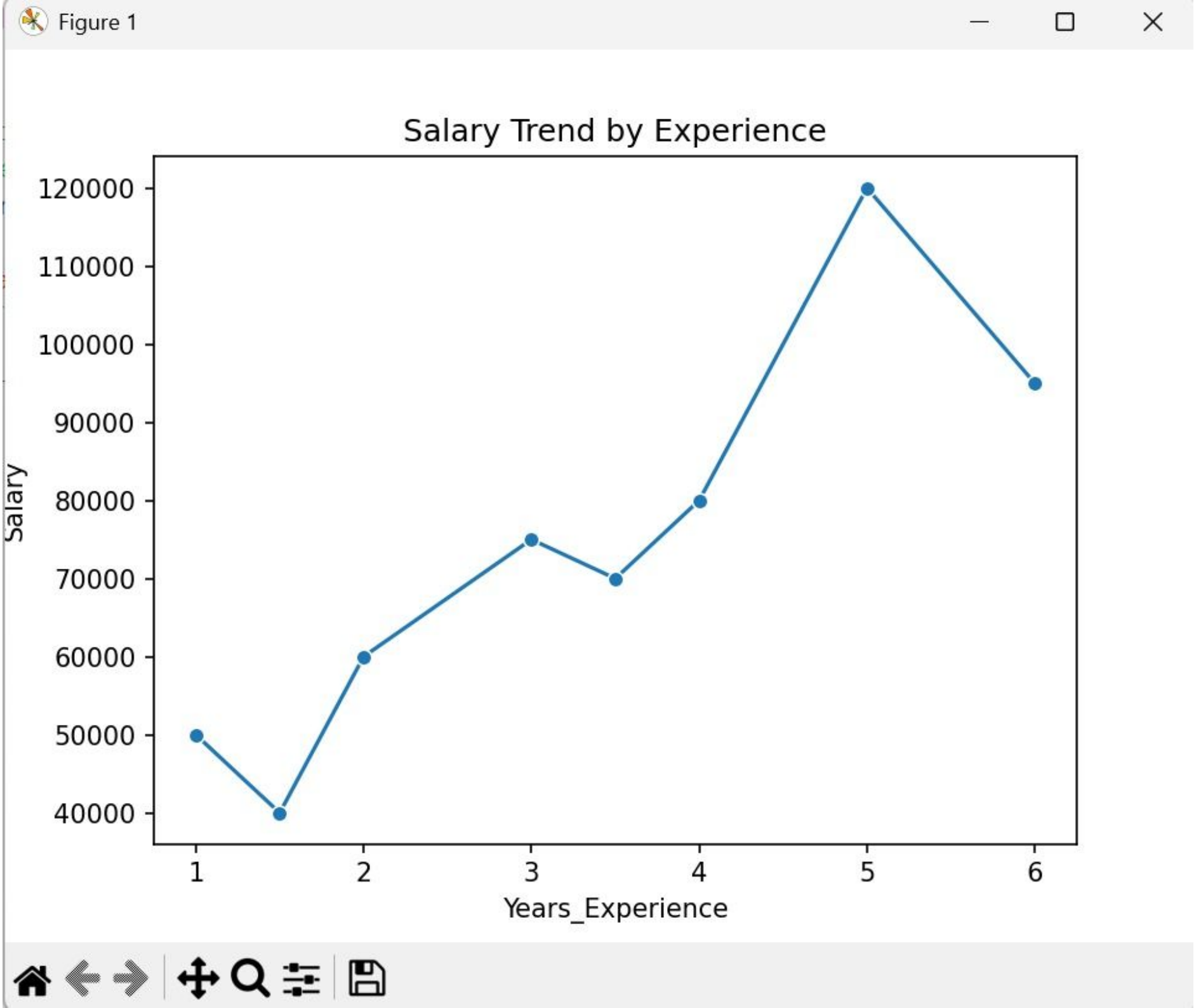
Figure 1
Salary Trend by Experience
120000
110000
100000
90000
80000
70000
60000
50000
40000
Salary
1
2
3
4
5
6
Years_Experience

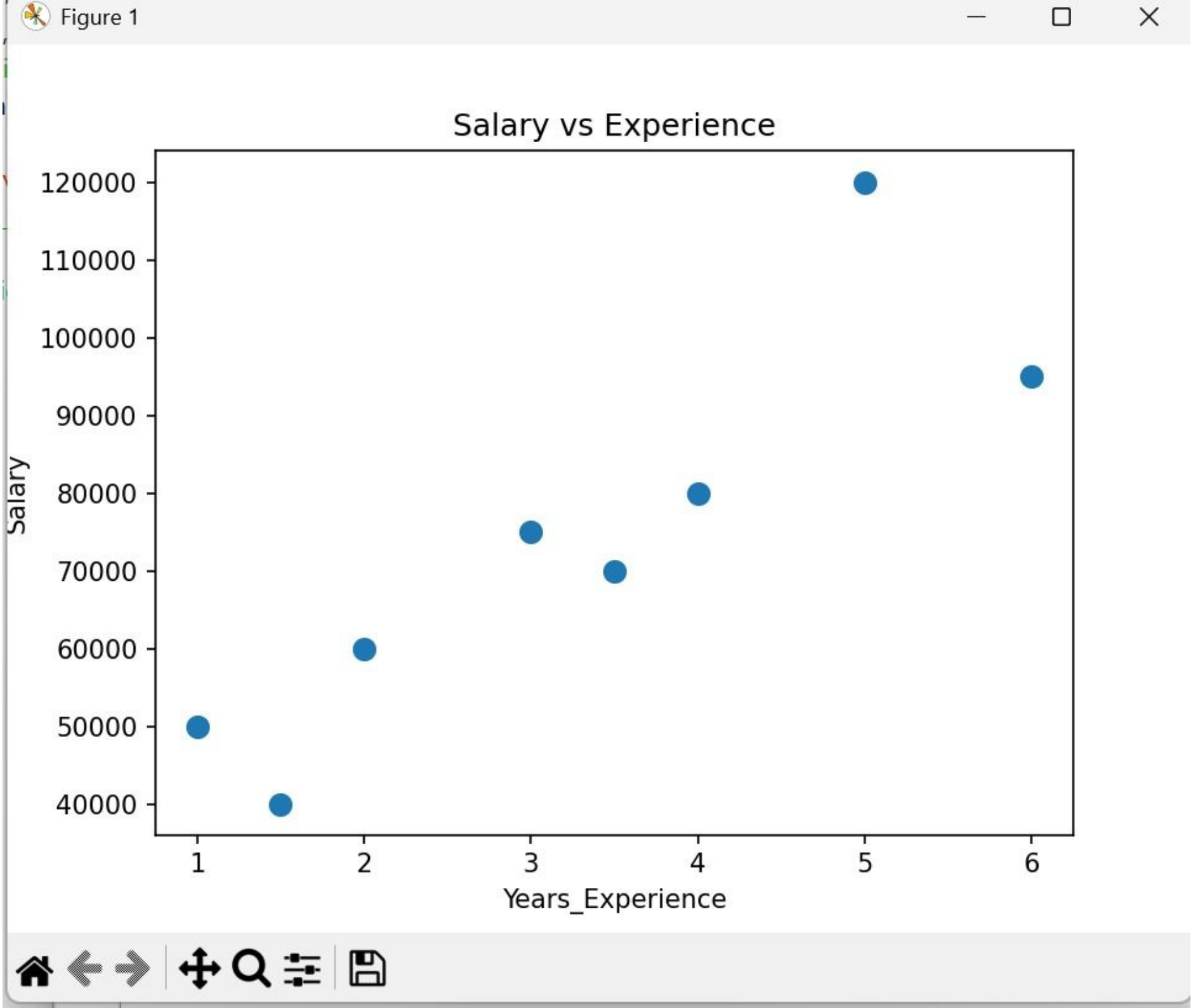
Figure 1
Salary vs Experience
120000
110000
100000
90000
80000
70000
60000
50000
40000
Salary
1
2
3
4
5
6
Years_Experience

Figure 1
Salary by Experience Level
120000
100000
80000
60000
40000
20000
0
Salary
Junior
Mid
Senior
Experience_Level